

ELEC9123: Design Task D

(Data Communication Networks)

Waiting Time Analysis for Markovian Queues Using Discrete Event Simulations

Term T2, 2026

1 Background

This **computer simulation**-based design project is aimed at understanding the various performance metrics in data communication networks experiencing random inter-arrival and service time for the users coming to the system. Further, to verify these performance metrics, Discrete Event Simulation (DES) needs to be performed for different system settings. You will use `MATLAB` or the mentioned computer simulations. Below, after briefly discussing the mentioned simulation settings, we discuss the adopted data network models.

2 Motivation

We all have experienced long waiting times in the queues for ticket booking or at payment terminals in supermarkets, airport baggage drop-off desks, and various other services. In all these queuing systems, the customers arrive randomly to obtain multiple services from the service providers. Before reaching the service desk(s), they have to wait for some random time for their turn in the queue(s) to get the desired service eventually. Here, service providers and customers waiting in the queues are generally interested in estimating the following quantities.

- The average number of customers waiting in the queue,
- The average delay experienced by each customer includes both the waiting time in the queue and the actual service time.

These quantities, as mentioned earlier, need to be estimated in terms of the available information on the customer inter-arrival rate and the typical service rate. Other than the mean values of these parameters, to predict average delay and queue length, more details on the statistical information regarding the distribution of the customer inter-arrival and service times are required. However, since these times are generally random, following the memoryless and independent distribution, the Markovian queues-based modelling can be used to find these parameters. Inter-arrival and service times are assumed to be exponentially distributed in such queuing systems.

This design task is targeted to one such application of Markovian queues for estimating *the waiting time and queue length in the airport bag-drop queues*. Later we extend it to more generic settings where customers arrive according to a Poisson process (inter-arrival times being exponentially distributed). Still, the customer service times have a general distribution-not necessarily exponential and can be deterministic too. Further, the Discrete Event Simulation (DES) is implemented in `MATLAB` to validate the corresponding waiting time and queue length analysis. In contrast to the conventional continuous simulation, a DES is governed by the discrete sequence of events in time. Each discrete event occurring at a particular time instant marks a change in the system's state. Moreover, as there is no change in the system's state between the concurrence of two consecutive events, this simulation jumps in time from one event to the next. So, the entire simulation is driven by the occurrence of successive events.

3 Problem Description

In the typical airport bag-drop queues, the passengers arrive randomly to drop their checked-in luggage(s), get their cabin bag(s) verified, and collect their boarding pass(es). Generally, there are limited service desks grouped under two categories; one for the business class (or priority) passengers and the other for the economy class (or normal) passengers.

Another special type of queuing system we notice frequently is where a maximum number of customers can be allowed. Specifically, the system can hold a total of K customers, including those in the service. Any further arriving customers will be refused entry to the design, and they will have to depart without any service thus immediately.

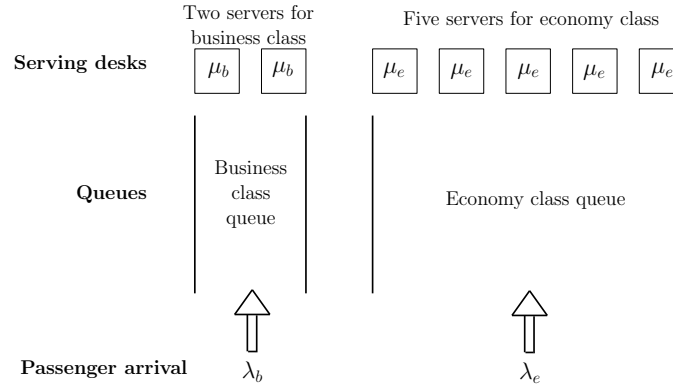


Figure 1: Two-class $M/M/m$ queuing system at the airport baggage-drop terminal.

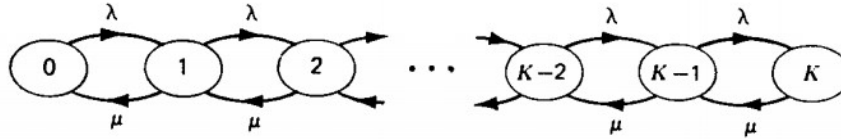


Figure 2: State-transition-rate diagram for finite storage room system $M/M/1/K$.

3.1 Introduction

In this design task, we will estimate the average queue length and the total time a passenger needs to spend in the queue and at the desk. As shown in Fig. 1, the airport baggage drop-off terminals generally have two queues. One with fewer counter/servers for the business class passengers and the other with more servers for the economy class. However, as the arrival rate of the economy class passengers is much faster owing to their much higher density or head counts, they face a much longer delay due to long queues because the number of servers is not as sufficient as in the case of the business class passengers.

Furthermore, to get a fundamental understanding of queuing systems and DES, we start with investigating and simulating the basic queues like $M/M/1$. After that, we will estimate the average system size, the time a customer needs to spend in the queue, and the service for a finite storage room queuing system $M/M/1/K$. Here the customers arriving with both inter-arrival times and their respective service times being exponentially distributed with rates λ and μ are served in a first-come, first-serve (FCFS) fashion by a single server. However, as shown in Fig. 2, a maximum number of K customers can be in the system simultaneously. Any further arriving customers beyond its capacity will be refused entry to the system and will depart immediately without service.

Other than basic queuing systems $M/M/1$, $M/M/1/K$ and two-class $M/M/m$ queuing system, we will also investigate $M/G/1$ queue where arrivals are Markovian (modulated by a Poisson process), service times have a General distribution, and there is a single server. Specifically, we have considered two general distributions:

- Uniformly distributed service time between $\frac{1}{2\mu}$ and $\frac{3}{2\mu}$, and
- deterministic service time $\frac{1}{\mu}$.

So, the goal of this design task is to learn and implement some smart tools that can be used to quantify the quality of experience (QoE) of the customers by estimating their average time spent for different queuing systems ($M/M/1$, $M/M/1/K$, $M/G/1$ and two-class $M/M/m$) and get insights on the average system size based on different arrival rates and the available service time for them. Also, since such systems are driven by the arrival and departure of the customers that occur as discrete events in time, this design task will be an exercise in learning this MATLAB based basic implementation of discrete-event simulation multiple queuing systems.

3.2 Presentation

This is one of the two Telecommunications based design tasks in the course ELEC9123. To pass the course, you need to solve this task according to the Tiered Learning taxonomy (TLT) guidelines and present it to a teacher at one of the lab exercises that offer examination. Explicitly following the TLT, the task is designed to provide an increasing complexity from Pass (Level 2) to High distinction (Level 5) levels as shown below:

- **Level 2 (Pass):** If you complete the correct implementation, understanding and explanation for $M/M/1$ system, the maximum mark you can achieve is 64 (Pass),

- **Level 3 (Credit):** If you complete the correct implementation, understanding and explanation for $M/M/1$ and $M/M/1/K$ systems, the maximum mark you can achieve is 74 (Credit),
- **Level 4 (Distinction):** If you complete the correct implementation, understanding and explanation for $M/M/1$, $M/M/1/K$, and $M/G/1$ (uniformly distributed service times and deterministic service time, i.e., $M/D/1$) systems, the maximum mark you can achieve is 84 (Distinction),
- **Level 5 (High Distinction):** If you complete the correct implementation, understanding and explanation for $M/M/1$, $M/M/1/K$, $M/G/1$, and the two class $M/M/m$ system, the maximum mark you can achieve is 100 (High Distinction).

The examination will be in English, so please write your code and comments in English (a good exercise since computer coding is common in companies). The problem should be individually, and you are not allowed to share or show MATLAB code or notes to other students. It is, however, permitted to discuss the design task orally with the course convener to get some good advice or clarification for any doubts!

- The entire design task can be divided into three sub-parts:
 - *Folders:* Depending upon which level to attempt based on the TLT framework for making/evaluation, you will make one to four folders, each containing the next main, analysis and simulation files. Specifically, for level 2, you will make a folder for $M/M/1$; for level 3, the folders will be on $M/M/1$ and $M/M/1/K$. Likewise for level 5, there will be 4 folders, namely, $M/M/1$, $M/M/1/K$, $M/G/1$, and $M/M/m$.
 - *Main file:* In this MATLAB file, you need to define all the variables and call the two functions that perform various tasks. Specifically, based on the type of queue passed in as input, depending on which level you decide to attempt, the right set of parameters should be passed to get the correct results returned by the respective functions. Further, all the plot generation and the comparison and validation of the results must be conducted here. Lastly, as a voluntary assignment, each main result can be exported to a text file using MATLAB commands.
 - *Analysis function:* This MATLAB function which is supplied by the known input parameters should return the average queue length and the average total time that each passenger spends at the baggage drop-off terminal by using the available closed-form expressions as presented later in Section 4.
 - *Simulation function:* You should write a MATLAB function DES that simulates a large occurrence of discrete events marked by the arrival and departure of the customers from the queuing system of interest. This function should be supplied with all the critical parameters like the maximum number of customers allowed, and the average arrival and service rates for all the cases. The output of this function should be the simulated average system length and their average service times. It can also be used for plotting the evolution of the queue with the occurrence of arrival/departure events.
- The solution should be demonstrated, and the code should be shown to the tutor. We will check if you have followed the guidelines, have a reasonable amount of comments in the code, if the plots are easy to understand, and if you can describe the different parts of the code. There is a style guide available on the course homepage, which elaborates on what a good solution should look like. Please make sure to read it and follow the recommendations!
- All design tasks must be conducted in MATLAB. The simulation files must be set up in a particular format as specified by each Design Task worksheet. The simulation file and a design log must be submitted via the Moodle course page.

3.3 Reference Material

Though most of the required details that should be sufficient for completing this design task are mentioned in this document itself, interested students are referred to read Chapters 3.3, 3.4 and 3.5 (especially Chapter 3.4.2) of the book [B1] (available online) for getting more insights on the Markovian queues discussed here.

More information on the basics of the $M/M/1/K$ queuing system can be obtained from Chapter 3.6 of [B2].

For further details on discrete event simulation, Chapter 2.2 (especially sub-chapter 2.2.2) of the book [B3] (also available online) can be referred to.

[B1] D. Bertsekas, R. Gallager, *Data Networks*, Second Edition, Prentice-Hall, vol. 15, issue 4, 1992. Available online: https://web.mit.edu/dimitrib/www/Queueing_Data_Nets.pdf

[B2] L. Kleinrock *Theory Volume 1 Queueing Systems*, Chichester U.K.:Wiley-Interscience 1975.

[B3] S. Robinson, *Simulation: The Practice of Model Development and Use*, John Wiley & Sons, 2004.

4 Theory and Analytical Results

In this section, we present the theory for the different queuing systems along with the average system size and time for analytical results that will be used for verifying the DES.

4.1 $M/M/1$ Queuing System

The $M/M/1$ queuing system represents a queuing station where the first letter in this three-letter notation indicates the nature of the arrival process with M denoting the memoryless, or generally a Poisson process, having exponentially distributed inter-arrival times for the customers in the system. The second letter indicates the nature of the probability distribution of the service times. Here again, M stands for the exponential service time distribution and the successive interarrival times and service times are assumed to be statistically independent of each other. The third letter 1 indicates only one server or service desk available for each customer type. Next, we present the analytical solutions for the $M/M/1$ system with Poisson distributed arrival processes and exponentially distributed service times.

The input parameters defining this queuing system are:

- Average arrival rate of customers: λ per unit time, and
- Average service rate for each customer: μ per unit time.

Hence, for $M/M/1$ system, the inter-arrival and service times are exponentially distributed with respective means $\frac{1}{\lambda}$ and $\frac{1}{\mu}$.

As in $M/M/1$ system, the average system size N representing the number people in the system, including the one in the service and others in the queue, for an $M/M/1$ queuing system is given by

$$N_{MM1} = \frac{\rho}{1 - \rho}, \quad (1)$$

where $\rho = \frac{\lambda}{\mu} < 1$ is the utilization factor. Finally, using the average system size expression (1) along with the average arrival rate and Little's theorem (Section 3.2.1 of book [B1]), the average time T_{MM1} spent by a person, including both waiting time in queue and actual service time, is given by

$$T_{MM1} = \frac{N_{MM1}}{\lambda} = \frac{1}{\mu - \lambda}. \quad (2)$$

4.2 $M/G/1$ queuing system

The $M/G/1$ queuing system represents a station where customers arrive according to a Poisson process with the rate λ . Still, the customer service times have a general distribution-not necessarily exponential. Using $\bar{X}^2$ to denote the second moment of service time and the Pollaczek-Khinchin (P-K) formula, the expected number of customers in the system N_{MG1} :

$$N_{MG1} = \rho + \frac{\lambda^2 \bar{X}^2}{2(1 - \rho)}, \quad (3)$$

where $\rho = \frac{\lambda}{\mu} < 1$ is the utilization factor. Here, it may be noted that the first moment of the mean of service time is $\frac{1}{\mu}$. Finally, using the average system size expression (3) along with the average arrival rate and Little's theorem, the average time T_{MG1} spent by a person, including both waiting time in the queue and actual service time, is given by:

$$T_{MG1} = \frac{N_{MG1}}{\lambda} = \frac{1}{\mu} + \frac{\lambda \bar{X}^2}{2(1 - \rho)}. \quad (4)$$

4.3 $M/M/1/K$ queuing system

The $M/M/1/K$ queuing system represents a queuing station with at most K customers allowed in the system. The last letter K represents the finite storage capacity for the $M/M/1/K$ system with Poisson distributed arrival processes and exponentially distributed service times. With maximum number of customers allowed in the system being K , we effectively turn off the Poisson input as soon as the system fills up. Therefore, the arrival and service rates for k th customer entering into the $M/M/1/K$ system are respectively defined as

$$\lambda_k = \begin{cases} \lambda, & k < K \\ 0, & k \geq K, \end{cases} \quad (5)$$

$$\mu_k = \mu, \quad k = 1, 2, 3, \dots, K. \quad (6)$$

The underlying probability of having k customers in the system at a given instant is given by

$$p_k = \begin{cases} \begin{cases} \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{K+1}} \left(\frac{\lambda}{\mu}\right)^k, & \lambda \neq \mu \\ \frac{1}{K+1}, & \lambda = \mu \end{cases}, & 0 \leq k \leq K \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

The average system size N_{MM1K} representing the number people in the system, including the one in the service and others in the queue, for an $M/M/1/K$ queuing system is given by

$$N_{MM1K} = \sum_{k=1}^K k p_k = \begin{cases} \frac{\lambda}{\mu} \left(\frac{1 - (K+1)\left(\frac{\lambda}{\mu}\right)^K + K\left(\frac{\lambda}{\mu}\right)^{K+1}}{\left(1 - \left(\frac{\lambda}{\mu}\right)^{K+1}\right)\left(1 - \frac{\lambda}{\mu}\right)} \right), & \lambda \neq \mu \\ \frac{K+1}{2}, & \lambda = \mu. \end{cases} \quad (8)$$

Finally, using the average system size expression (8) along with the average arrival rate and Little's theorem, the average time T_{MM1K} spent by a person, including both waiting time in queue and actual service time, is given by

$$T_{MM1K} = \frac{N_{MM1K}}{\bar{\lambda}} = \begin{cases} \frac{1}{\mu} \left(\frac{1 - (K+1)\left(\frac{\lambda}{\mu}\right)^K + K\left(\frac{\lambda}{\mu}\right)^{K+1}}{\left(1 - \left(\frac{\lambda}{\mu}\right)^K\right)\left(1 - \frac{\lambda}{\mu}\right)} \right), & \lambda \neq \mu \\ \frac{K+1}{2\lambda}, & \lambda = \mu, \end{cases} \quad (9)$$

$$\text{where the average arrival rate } \bar{\lambda} = \sum_{k=0}^K \lambda_k p_k = \begin{cases} \lambda \left(\frac{1 - \left(\frac{\lambda}{\mu}\right)^K}{1 - \left(\frac{\lambda}{\mu}\right)^{K+1}} \right), & \lambda \neq \mu \\ \lambda \left(\frac{K}{K+1} \right), & \lambda = \mu. \end{cases}$$

4.4 $M/M/m$ queuing system

The $M/M/m$ queuing system represents a queuing station with a m servers. The first letter in this three-letter notation indicates the nature of the arrival process with M denoting the memoryless, or in general a Poisson process, having exponentially distributed inter-arrival times for the customers in the system. The second letter indicates the nature of the probability distribution of the service times. Here again, M stands for the exponential service time distribution and the successive interarrival times and service times are assumed to be statistically independent of each other. The last letter m indicates the number of servers or service desks available for each passenger type.

With the number of servers being m , the average queue length N_Q for an $M/M/m$ queuing system is given by

$$N_Q \triangleq \frac{p_0 \rho (m\rho)^m}{m! (1 - \rho)^2}, \quad (10)$$

where p_0 representing that there are no customers in the system is defined as

$$p_0 \triangleq \left[\sum_{i=0}^{m-1} \frac{(m\rho)^i}{i!} + \frac{(m\rho)^m}{m! (1 - \rho)} \right]^{-1}, \quad (11)$$

and the utilization factor ρ of this queuing system, representing the long-term proportion of time the server is busy, is defined as

$$\rho \triangleq \frac{\lambda}{m\mu} < 1. \quad (12)$$

Using the average queue length expression (10) along with the average arrival rate and Little's theorem, the average waiting time T_Q in queue is given by

$$T_Q \triangleq \frac{N_Q}{\lambda}. \quad (13)$$

Finally, using (13), the average total time T_{MMm} spent by each customer in the system is given by

$$T_{MMm} \triangleq T_Q + \frac{1}{\mu}. \quad (14)$$

5 Suggested Solution Outline

Begin by reading through the whole document so that you understand the problem specification and our suggested way to divide the problem into smaller subproblems.

The steps below provide one way to split the main programming problem into subproblems that can be solved one at a time. We have chosen the subproblems to make it possible to verify each part before you continue with the next part. Make sure to take these opportunities other than the TLT evaluation framework!

Although the solution outline suggests that you write small functions that solve subproblems, this doesn't mean that it is good or efficient to use (call) all these small functions to solve the original problem. Some functions take care of such small pieces that it is better to copy the code into a new larger function than to call the small function.

It is worth noting that this solution outline is intended for students with no or little previous programming experience. If you consider yourself an experienced programmer and think you have a better approach to solving the problem, you are allowed to solve it in your way. An experienced programmer would probably, based on previous experience, divide the problem into different and/or fewer parts. However, if you need help, the teaching assistant has more understanding of the outlined solution. If you don't follow the solution outline, you must ensure that the solution gives the same functionality.

5.1 Initializing the System Parameters

5.1.1 $M/M/1$ and $M/G/1$ Systems

For the basic queues like $M/M/1$ and $M/G/1$, the input parameters are:

- Average arrival rate of customers: $\lambda = 10$ customers/hour, and
- Average service time for a customer: $\frac{1}{\mu} = 5$ minutes.

Here, the service time is exponentially distributed with a mean of 5 minutes for $M/M/1$, uniformly distributed between $a = 2.5$ and $b = 7.5$ minutes for $M/G/1$ with a mean of 5 minutes and $\overline{X^2} = \frac{a^2+ab+b^2}{3}$. For the $M/D/1$ queue, the service time is deterministic/constant with a value of 5 minutes and $\overline{X^2} = 5^2 = 25$. The analytical expressions for average system sizes are given by (1) and (3), for $M/M/1$ and $M/G/1$, respectively. The underlying average time spent in the system can be obtained by substituting their respective system parameters in equations (2) and (4).

5.1.2 $M/M/1/K$ system: Finite Storage

In $M/M/1/K$, the customers arriving with both inter-arrival times and their respective service times being exponentially distributed with rates λ and μ are served in a first-come, first-serve (FCFS) fashion by a single server. However, as shown in Fig. 2, a maximum number of K customers can be in the system at a time. Any further arriving customers beyond its capacity will be refused entry to the system and will depart immediately without service.

The input parameters defining this queuing system are:

- Storage capacity $K = 10$
- Average arrival rate of customers: $\lambda = 10$ customers/hour
- Average service time for a customer: $\frac{1}{\mu} = \begin{cases} 5 \text{ minutes,} & \text{case 1} \\ 6 \text{ minutes,} & \text{case 2} \end{cases}$

The average system size N_{MM1K} and the average service time T_{MM1K} for the $M/M/1/K$ system can be obtained by substituting their respective system parameters in equations (8) and (9), respectively.

5.1.3 Two-class $M/M/m$ system: Business and Economy classes

Lastly, we investigate a two-class (business and economy) $M/M/m$ queuing system, with Poisson distributed arrival processes and exponentially distributed service times for each queuing system. So, as also depicted via Fig. 1, there are two independent queuing systems, one dedicated to the business class passengers and the other for servicing the economy class passengers. The input parameters defining this queuing system at the airport baggage drop-off desks are:

- Number of desks for the business class: $m_b = 2$
- Number of desks for economy class: $m_e = 5$
- Average arrival rate of business class: $\lambda_b = 20$ passengers/hour

- Average arrival rate of economy class: $\lambda_e = 100$ passengers/hour
- Average service time for a business class passenger: $\frac{1}{\mu_b} = 2.5$ minutes
- Average service time for a economy class passenger: $\frac{1}{\mu_e} = 2.5$ minutes

The average queue length and service time for the business and economy class passengers can be obtained by substituting their respective system parameters in equations (10) and (14), respectively.

5.2 Discrete Event Simulation: Basic Insights and Hints

Next, using MATLAB implement the discrete event simulation (DES) for the above two-class queuing system. This implementation involves the following steps.

- Write a function that runs the discrete event simulation (DES) and returns the simulated average queue length and total time in the system of interest ($M/M/1$, $M/G/1$, $M/M/1/K$, and Two-class $M/M/m$) for a given set of input parameters, i.e., the system size, number of servers along with the arrival and service rates.
- **Note 1:** Students are **not permitted to use in-built commands or the simulink tools** for implementing the mentioned Markov queues. You must write your own routines and programs for implementing these queues.
- Here, it may be noted for $M/M/-/-$ queuing systems, both inter-arrival and service times following exponential distribution can be generated using the in-built MATLAB function `expnrnd` with its parameter being respectively the reciprocal of the mean inter-arrival time or mean service time. Whereas for the $M/G/1$ system, the service time can be generated using the in-built MATLAB function `rand` for uniformly distributed service times and kept as constant service time for deterministic service times. So, these random variables will receptively represent the time between two consecutive passenger arrivals and the service time for a passenger.
- In the simulation file, first define the simulation `runtime`. For example, to set $y = 6$ months, initialize `runtime = y × 30 × 24 × 60` (in minutes).
- *Variables that can be included are to keep track of system evolution are:*
 - the counting variable for recording the changes in the system's state, either due to the arrival or departure of a passenger.
 - One dimensional array for storing the arrival and departure times of customers (business and economy class passengers in case of two-class $M/M/m$ queuing system). These can be later used for calculating the total time (waiting in queue + basic service) spent by each passenger in the system.
 - A variable for keeping track of the current time which in turn also marks the occurrence of each event in time along with the termination of the simulation when the maximum `runtime` is reached.
 - One dimensional array for keeping track of the variation of the count of customers (both business and economy class passengers for two-class $M/M/m$ system) at each discrete event's occurrence
 - Likewise, the state of each queue, i.e., the length of both business and economy class queues, also needs to be stored to obtain the simulated average queue lengths later.
- *Initialization, iteration, and termination*
 - For $M/M/1$, $M/M/1/K$, $M/G/1$ systems
 - * Start the DES by setting the time $t = 0$.
 - * The simulation time t then jumps to the time of occurrence of the first arrival of a customer, as obtained by selecting the minimum among the exponentially distributed random inter-arrival times with mean $\frac{1}{\lambda}$.
 - * This process of generating the exponentially distributed random inter-arrival and service times (note: for $M/G/1$ service time is either uniformly distributed or deterministic in this task) repeatedly continues till the simulation is terminated with the time $t = \text{runtime}$ reached.
 - * Each step in the above-mentioned iterative process is governed by the occurrence of either a customer's arrival or departure and the corresponding changes in the queue and the overall system.
 - For two-class $M/M/m$ system
 - * Start the DES by setting the time $t = 0$.
 - * The simulation time t then jumps to the time of occurrence of the first arrival of a passenger, either business or economy class, as obtained by selecting the minimum among the exponentially distributed random inter-arrival times with respective means $\frac{1}{\lambda_b}$ and $\frac{1}{\lambda_e}$.

- * This process of generating the exponentially distributed random inter-arrival and service times continues until the simulation is terminated with the time $t = \text{runtime}$ reached.
- * Each step in the above-mentioned iterative process is governed by the occurrence of a passenger's arrival or departure and the corresponding changes in the two queues and the overall system.
- *Key logic to be implemented during the DES*
 - For $M/M/1$, $M/M/1/K$, $M/G/1$ systems
 - * The DES is governed by the occurrence of discrete events like the arrival and departure of customers.
 - * So, the system's state needs to be investigated at each of these two events, namely,
 - arrival of a customer,
 - departure of a customer.
 - * The arrival of a customer is followed by their service at the desk or joining the queue if the server is busy and the number of customers in the system including it is $\leq K$. However, for $M/M/1$ and $M/G/1$ queues, there is no such restriction on system size. The next arrival and service times are obtained by generating the exponentially distributed random variables with their respective means for $M/M/1$ and $M/M/1/K$. Whereas for $M/G/1$, though inter-arrival time is exponentially distributed, the service times are either considered uniformly distributed or deterministic with the same mean $\frac{1}{\mu}$.
 - * Whereas, a customer's departure is followed by their departure from the service desk. The first customer in the queue gets the service now with their time spent at the service desk obtained by generating another exponentially distributed random variable for $M/M/1$ or $M/M/1/K$ (or as uniformly distributed one for $M/G/1$ or constant for deterministic $M/D/1$) having to mean equivalent to the average service time, i.e., $\frac{1}{\mu}$. However, if there is no customer in the queue, the next departure time should be set as *infinity* or a very high value.
 - * Store the updates in both the queues and the overall system state at the arrival and departure of each customer.
 - * Only for $M/M/1/K$, if on the arrival of a new customer, the total customer count becomes K , then the next arrival time should be set as *infinity* or a very high value.
 - * Store the updates in both the queues and the overall system state at the arrival and departure of each customer.
 - * This stored evolution of the queue state with the arrival/departure events and the arrival and departure times of each customer is used to obtain later the simulated average system size and service time for customers.
 - For two-class $M/M/m$ system
 - * The DES is governed by the occurrence of discrete events like the arrival and departure of the business and economy passengers.
 - * So, the system's state needs to be investigated at each of these four events, namely,
 - arrival of a business class passenger,
 - departure of a business class passenger,
 - arrival of an economy class passenger, and
 - departure of an economy class passenger.
 - * The arrival of a passenger is followed by either their service at the desk or joining the queue if all servers are busy. The next arrival and service times are obtained by generating the exponentially distributed random variables with their respective means.
 - * Whereas, the departure of a passenger is followed by their departure from the service desk, and the first passenger in the queue gets the service now with their time spent at the service desk obtained by generating another exponentially distributed random variable having to mean equivalent to the average service time, i.e., $\frac{1}{\mu_b}$ or $\frac{1}{\mu_e}$. However, if there is no passenger in the queue, the next departure time should be set as *infinity* or a very high value.
 - * Store the updates in both the queues and the overall system state at the arrival and departure of each passenger.

- * This stored evolution of the queue state with arrival/departure events and the arrival and departure times of each passenger is used to obtain later the simulated average queue length and service time for both business and economy class passengers.

5.3 Intermediate Key Checks and Alternate Suggestions

- The exponentially distributed inter-arrival and service times can also be generated collectively in one go as two random vectors, with each having size equal to the number of discrete events for arrival and departure that need to be simulated.
- The distribution of the inter-arrival and service times can be validate using the `ecdf` function in MATLAB and comparing the corresponding CDF with the analytical one as given by $F(x) = 1 - e^{-\lambda x}$. Here $\frac{1}{\lambda}$ is the mean of the underlying exponentially distributed random variable.
- For the $M/G/1$ case with service times being uniformly distributed, `rand` function in MATLAB can be directly used.
- It should be noted that the results for the average system size and time spent in the $M/M/1/K$ system are different for $\lambda = \mu$ and $\lambda \neq \mu$.

5.4 Compute Estimates of the Simulated Expected Values

At the end of the simulation run, this `DES` function should also return the estimated value of the system size N and the total time T spent in the system by each customer. Further, as the events occur at discrete times, it is essential to carefully take the time duration between each consecutive event while obtaining the time average system size.

5.5 Compare the Simulated Results with Analytic Solutions

After collecting the results from the analysis and simulation files in the main file, plot them via a bar chart to compare their respected values. Section 4 presented the theoretical expected values. Make sure that the estimates from the simulation function are close to the exact analytical values!

Remember that, if you attempt for Level 3 or above, compare the performance of different queuing systems: $M/M/1$, $M/M/1/K$, $M/G/1$, $M/D/1$, and two-class $M/M/m$.

Further, obtain the percentage deviation of the simulated results from the ones obtained using the closed-form expressions defined in Section 4. Also, check if this percentage error or gap can be reduced by increasing `runtime`.

6 Useful Facts

- For the system stability, the utilization factor $\rho < 1$, i.e., $\lambda < m\mu$ for single server queues and $\lambda < m\mu$ for $M/M/m$ queuing system. If $\rho \geq 1$, then server(s) cannot keep up with the arrival rate (which is larger than the service rate) and the queue length increases without bound, causing the system to become unstable. For more details, please refer to [B1, pages 168-169].
- The simulation time should be sufficiently long to ensure that at least about 10^4 events (arrival or departure) are considered for obtaining the simulated results, which are closely matched to those obtained analytically.
- Random number for any distribution can be generated using the `rand` function in MATLAB for generating the uniform random variable between 0 and 1. The trick is based on the cumulative distribution function (CDF) also lies in the range between 0 and 1. The formula for generating the random variable x having the CDF F is given by $x = F^{-1}(\text{rand})$.
- For example with CDF $F(x) = 1 - e^{-\lambda x}$ of exponentially distributed random variable having mean $\frac{1}{\lambda}$, the formula for generating the exponentially distributed random variable x is given by $x = -\frac{1}{\lambda} \log(\text{rand})$.
- The number of random samples to be generated for simulation should be sufficiently large to ensure that at least about 10^6 are considered for obtaining the simulated results, which are closely matched to those obtained analytically. The improvement in accuracy can be observed by increasing the number of random samples to 10^7 .
- Lastly, for attempting Level 5 and implementing two-class $M/M/m$ queues, note that, generally, the number of servers for the economy class is not as high as their arrival rate. Mostly, $\lambda_e \approx 5\lambda_b$. At the same time, their respective service rates are almost the same. So, the business class passengers generally have to face less waiting time than the economy class.

7 TLT Assessment Structure

To pass the course, you need to solve this task according to the **Tiered Learning taxonomy (TLT)** guidelines and present it to an instructor/demo at one of the lab exercises that offer examination. If you choose to do designs above Level 2, you should follow them in a sequential order. For instance, you should finish Level 2 before trying Level 3 and so on. The highest score you can get is based on the level you choose. Explicitly following the TLT, this design task is designed to provide an increasing complexity from Pass (Level 2) to High Distinction (Level 5) levels, as shown below:

- If you complete the design task under Level 2 successfully with the correct implementation of the $M/M/1$ system, the maximum mark you can achieve is 64 (Pass)
- If you complete the design task under Level 3 successfully with the correct implementation of $M/M/1$ and $M/M/1/K$ systems, the maximum mark you can achieve is 74 (Credit)
- If you complete the design task under Level 4 successfully with the correct implementation of $M/M/1$, $M/M/1/K$, and $M/G/1$ systems, the maximum mark you can achieve is 84 (Distinction)
- If you complete the design task under Level 5 successfully with the correct implementation of $M/M/1$, $M/M/1/K$, $M/G/1$, and the two class $M/M/m$ systems, the maximum mark you can achieve is 100 (High Distinction)

8 Attendance Requirements

- Student attendance in each laboratory session (Wednesday and Friday) will be **recorded**.
- In order to receive a grade for this design task, it is **mandatory** to attend all required lectures/labs, including the lab assessments.
- To ensure consistency and fairness among different simulation and hardware-focused students, all the lab-based questions will be answered or addressed only during the two three-hour weekly lab sessions.

9 Design Journal Submission

- It is mandatory to maintain a journal or report describing your design process, implementation code, and results, which must be documented via an MS Word or .pdf file.
- Please ensure that you take photos or record the completion of each level as you move to the next so that you have sufficient implementation details and results to be documented in your design journal or logbook.
- All design tasks in this course must be completed individually. Copying from others is not permitted and may result in a failing grade, reinforcing the importance of academic integrity.
- The design journal should include a functioning and well-commented MATLAB file containing your design that meets the requirements of the design task and must be submitted to the **Moodle Design Task X submission link**, where X is 1, 2, or 3 depending on whether you chose this task as your first, second, or third design task.

Please ensure that the MATLAB simulation file is formatted according to the requirements listed in Section 5.

- During the assessment, you will also need to demonstrate the functionality of the MATLAB code, explain your design journal and your design to the lab demonstrator and answer any questions they may have about your design.
- Marking of this task will include the demonstration of the task and the evaluation of the lab notebook/report/journal.
- Your journal should already have all the subparts required to showcase the necessary functionality, along with detailed comments explaining all the steps. You may be penalised if significant additions or changes are required to show functionality or demonstration.
- **Submission File Format**
 - The simulation file must be saved as a '.m' file (MATLAB file). The name of this file must be: `zID_LastName_FirstName_DTD_2026.m`. Here if any student has written their code in multiple MATLAB files, then they can compress all the .m together in a single zip folder while exactly mentioning which file to run and what is the purpose of each .m file by adding a README text file explaining the role of each .m file. The name of this zip folder, in this case, must be `zID_LastName_FirstName_DTD_2026.zip`
 - Mention the attempted Tiered Learning Taxonomy (TLT) level so you can be evaluated accordingly.

- Your simulation file should already have all the subparts required to showcase the necessary functionality, along with detailed comments explaining all the steps. You may be penalized if significant additions or changes are required to show functionality per the mentioned requirements.
- The brief portable document format (pdf) report summarizing the outcomes of this Design task along with your steps in obtaining them must be saved as a '.pdf' file (readable text file). The name of this file must be zID_LastName_FirstName_DTD_2026.pdf
- All submissions for this course will be subject to **plagiarism and similarity checking using Turnitin**. In addition, submissions may be reviewed using **AI detection and academic integrity assessment tools** to identify potential unauthorised or inappropriate use of generative AI systems. So, the students are expected to ensure that all submitted work is their own original work and that any external sources, references, code snippets, online materials, or AI-assisted contributions are appropriately acknowledged and declared.

10 Design Task Schedule and Assessment Timeline

- Each design task must be completed over a span of six lab sessions. Term 2 is structured to accommodate three design tasks to be selected by each student out of the 9 available options, with the following schedule:
 - Task 1: Week 1 Wednesday lab session to Week 3 Friday lab session,
 - Task 2: Week 4 Friday lab session to Week 7 Wednesday lab session,
 - Task 3: Week 8 Wednesday lab session to Week 10 Friday lab session.
- Each task has a specific assessment session and submission deadline:
 - If this task is completed as Task 1:
 - * Assessment: Wednesday lab session of Week 4,
 - * Deadline: 11:59 PM, Tuesday, June 23, 2026.
 - If this task is completed as Task 2:
 - * Assessment: Friday lab session of Week 7,
 - * Deadline: 11:59 PM, Thursday, July 16, 2026.
 - If this task is completed as Task 3:
 - * Assessment: Wednesday lab session of Week 11,
 - * Deadline: 11:59 PM, Tuesday, August 11, 2026.

11 Assessment Criteria and Marking Rubric

- The total mark for the assignment is determined from the following submissions:
 - A functioning and well-commented MATLAB file containing your design that meets the requirements of the design task must be submitted to the Moodle Design Task 2 submission link. You will need to demonstrate the functionality of the MATLAB code to the lab demonstrators. Please ensure that the MATLAB simulation file is formatted according to the requirements listed in Section 5.
 - As mentioned earlier, the task is designed to provide an increasing complexity from Pass (Level 2) to High Distinction (Level 5) levels, as shown below: Under the TLT framework, the maximum marks (design code + the design journal/report) you can achieve are as follows:
 - * Level 2 (Pass): Correct implementation, understanding and explanation of the $M/M/1$ system
 - * Level 3 (credit): Correct implementation, understanding and explanation of $M/M/1$ and $M/M/1/K$ systems
 - * Level 4 (Distinction): Correct implementation, understanding and explanation of $M/M/1$, $M/M/1/K$, and $M/G/1$ systems
 - * Level 5 (High Distinction): Correct implementation, understanding and explanation of $M/M/1$, $M/M/1/K$, $M/G/1$, and the two class $M/M/m$ systems
- The assessment breakdown for each design task, showing different components involved, is shown in Table 1.
- The Correct Working Design component, worth 20%, requires that **all objectives are accurately met** according to the selected level, as described in the first bullet of this section.

Table 1: The assessment breakdown for Design Task D

Deliverable	Percentage of Assignment Grade
Demonstration of Correct Working Design	20%
Explanation of the Implementation and Results	40%
Ability to Answer Questions	20%
Design Journal or Notebook	20%

- The Explanation of Implementation and Results is evaluated based on the following four aspects:
 - **Clarity and explanation of plots** - how well the plots are presented and interpreted.
 - **Code efficiency and commenting** - the quality, readability, and documentation of the code.
 - **Working code and its explanation** - whether the code runs correctly and is clearly explained.
 - **Explanation of Results** - how well the outcomes are analysed and related to the design objectives.
- The "Ability to Answer Questions" component is based on the quality of responses to the **two to four questions** based on the design task.
- The Design Journal component is evaluated based on four equally weighted areas. First, students must provide a brief **problem description** that clearly outlines the design task. Second, they should present their **understanding and logic**, explaining their approach and reasoning in a structured and coherent manner. Third, the journal should include a demonstration of results, showcasing the outcomes of their design work with appropriate interpretation and validation of accuracy. Finally, students must include **project management** details, such as weekly progress updates, a declaration of any AI tools used, and proper referencing of all sources.
- Overall, during the assessment, students will need to demonstrate the functionality of the MATLAB code, explain their design journal and design to the lab demonstrator, and answer any questions they may have about their design.
- All design tasks in this course must be completed **individually**. Copying from others is not permitted and may result in a failing grade, reinforcing the importance of academic integrity.
- Please note that design tasks must be submitted by the set deadline, and extensions will not be granted, except in cases of approved special consideration, which may allow students to redo the task during Weeks 11 and 12.
- **Passing Requirement and Hurdle:** To pass this course, you must:
 - Achieve an overall course mark of 50% or higher, where this overall mark is calculated as follows:
 - * 30% from each of the three design tasks (totaling 90%),
 - * 10% from the lab exam, which will be conducted during the Friday lab session of Week 11, and
 - Pass at least two out of your three design tasks. This means you can fail only one design task.
- **Redo Opportunity:** If you fail a design task, you will have a chance to redo it during Weeks 11 and 12.
 - Lab sessions for redoing tasks will run from 9:00 AM to 3:00 PM on:
 - * 6-hour lab session on Wednesday of Week 11
 - * 6-hour lab sessions on Wednesday and Friday of Week 12
 - If your resubmitted task meets the requirements for a satisfactory grade, you will receive a mark of 50% for that task.
 - The grading and assessment will be conducted on the Friday lab session of Week 12 after the student submits their task during the same session.
 - Please be aware that you are allowed to **redo ONLY one design task**.
- **Supplementary Lab Sessions:** The above-mentioned three 6-hour lab sessions in Weeks 11 and 12 can also be used as make-up sessions for students who missed earlier labs due to approved special consideration. In such cases, students will be assessed normally and will receive their actual earned mark, rather than the fixed 50% awarded for task redos. Please be aware that students are strictly permitted to complete only one design task during the supplementary lab sessions in Weeks 11 and 12. No exceptions will be made beyond this single opportunity.
- If needed, students may be given access to open lab sessions to work outside regular lab hours. However, lecturer and demonstrator support will only be available during the two scheduled weekly lab sessions.

- **Formal feedback** will be provided well within two weeks of the relevant submission date through Moodle, followed by a brief discussion during the lab session.
- **Note:** Use of AI tools (such as ChatGPT or Copilot) for your learning in this course is allowed, but you are responsible for everything you produce - designs, reports, graphs, etc. (keeping in mind that the output from AI tools can be incorrect). Any use of AI tools must be declared.